

练习二十 静电场 (二)

1. 0

2. $\frac{q}{4\pi\epsilon_0 r} - \frac{q}{4\pi\epsilon_0 R}$

3. (A)

4. (2)

5. $U = \int_0^R \frac{\sigma 2\pi r dr}{4\pi\epsilon_0 (r^2 + x^2)^{1/2}} = \frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + x^2} - x);$

$$E = -\frac{dU}{dx} = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{R^2 + x^2}}\right)$$

6. (1) $\frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3}\right);$

(2) $\frac{Q^2}{8\pi\epsilon_0 R_3}$